(a) Tracking the distributions of newly emerged interactions when we injected more non-generalizable representations **to DNN's parameters.**

(b) Tracking the distributions of newly emerged interactions when we injected more non-generalizable representations **to input samples.**

Increasing noise magnitude σ

Increasing noise magnitude σ

$\sigma = 0.01$ $\sigma = 0.02$ $\sigma = 0.03$ $\sigma = 0.04$ $\sigma = 1/255$ $\sigma = 2/255$ $\sigma = 3/255$ $\sigma = 4/255$

ResNet-20 on MNIST

ResNet-20 on CIFAR-10

VGG-11 on CIFAR-10

$\sigma = 1\times10^{-8}$ $\sigma = 3\times10^{-8}$ $\sigma = 1\times10^{-7}$ $\sigma = 3\times10^{-7}$ $\sigma = 1\times10^{-8}$ $\sigma = 3\times10^{-8}$ $\sigma = 1\times10^{-7}$ $\sigma = 3\times10^{-7}$

Qwen2.5-7B on Finance

Llama-3-8B on Finance

GLM-Z1-9B on Finance

Phi-4-14B on Finance

Order m

Empirically measured newly emerged interaction distribution $\Delta A^{(m),+}$ $\Delta A^{(m),-}$

Theoretically derived spindle-shaped distribution $\mathcal{A}^{(m),+}_{\text{spindle}}$ $\mathcal{A}^{(m),-}_{\text{spindle}}$